grading, abstaining is strictly sub-optimal. IDK-type responses are maximally penalized while an overconfident "best guess" is optimal. The motivation combines two desirable factors: (a) the rate of accuracy among what is output by the language model, and (b) how comprehensive responses are. However, weighing (a) more than (b) is important for reducing hallucinations.

Formally, for any given question in the form of a prompt c, denote the set of plausible responses (valid or error) by $\mathcal{R}_c := \{r \mid (c, r) \in \mathcal{X}\}$. Further, suppose there is a set of plausible abstention responses $\mathcal{A}_c \subset \mathcal{R}_c$ (e.g., IDK). A *grader* $g_c : \mathcal{R}_c \to \mathbb{R}$ is said to be *binary* if $\{g_c(r) \mid r \in \mathcal{R}_c\} = \{0, 1\}$ and $g_c(r) = 0$ for all $r \in \mathcal{A}_c$. A *problem* is defined by $(c, \mathcal{R}_c, \mathcal{A}_c, g_c)$ where the test-taker knows $c, \mathcal{R}_c, \mathcal{A}_c$. We assume that the test-taker knows that the rubric is binary but is not told the correct answers, where $g_c(r) = 1$. The test-taker's beliefs about the correct answer can be viewed as a posterior distribution ρ_c over binary g_c's. For any such beliefs, the optimal response is not to abstain.

Observation 1. *Let c be a prompt. For any distribution ρ_c over binary graders, the optimal response(s) are not abstentions, i.e.,*

$$\mathcal{A}_c \cap \arg\max_{r \in \mathcal{R}_c} \mathop{\mathbb{E}}_{g_c \sim \rho_c}[g_c(r)] = \emptyset.$$

Although the proof is trivial (see Section E), Observation 1 suggests that *existing evaluations may need to be modified.* Table 2 summarizes the short meta-evaluation analysis in Section F, finding that the vast majority of popular evaluations have binary grading. Therefore, additional hallucination evaluations may not suffice when the primary evaluations penalize honestly reporting confidence and uncertainty. This does not diminish existing work on hallucination evaluations but rather points out that even the ideal hallucination evaluation and ideal post-training methodology, yielding honest reports of uncertainty, may still be drowned out because of inferior performance on the vast majority of the existing evaluations.

4.2 Explicit confidence targets

Human tests are similarly mostly binary, and it has been recognized that they also reward overconfident bluffing. Of course, exams are only a small component of human learning, e.g., fabricating birthdays will quickly result in embarrassment. Nonetheless, some standardized national exams operate or have operated using penalties for incorrect answers (or equivalently partial credit for abstaining), including Indian JEE, NEET, and GATE exams; AMC tests from the Mathematical Association of America; and US standardized SAT, AP, and GRE tests in earlier years. Importantly, the grading system is clearly stated in the instructions, and test takers are often aware of the confidence threshold beyond which it makes sense to make their best guess.

Similarly, we propose evaluations explicitly state *confidence targets* in their instructions, within the prompt (or system message). For example, one could append a statement like the following to each question:

> Answer only if you are $> t$ confident, since mistakes are penalized $t/(1-t)$ points, while correct answers receive 1 point, and an answer of "I don't know" receives 0 points.

There are several natural values of t including $t = 0.5$ (penalty 1), $t = 0.75$ (penalty 2), and $t = 0.9$ (penalty 9). A threshold of $t = 0$ corresponds to binary grading and could be described by, e.g., "Make your best guess even if you are unsure, as if you were taking an exam." A simple calculation

Table 2: Summary of evaluation benchmarks analyzed in this work and their treatment of abstentions. "Binary grading" indicates that the primary metric is a strict correct/incorrect accuracy; "IDK credit" denotes whether abstentions can earn any credit.

Benchmark	**Scoring method**	**Binary grading**	**IDK credit**
GPQA	Multiple-choice accuracy	Yes	None
MMLU-Pro	Multiple-choice accuracy	Yes	None
IFEval	Programmatic instruction verification	Yes[a]	None
Omni-MATH	Equivalence grading*	Yes	None
WildBench	LM-graded rubric*	No	Partial[b]
BBH	Multiple-choice / exact-match	Yes	None
MATH (L5 split)	Equivalence grading*	Yes	None
MuSR	Multiple-choice accuracy	Yes	None
SWE-bench	Patch passes unit tests	Yes	None
HLE	Multiple-choice / equivalence grading*	Yes	None

* Grading is performed using language models, hence incorrect *bluffs* may occasionally be scored as correct.
[a] IFEval aggregates several binary rubric sub-scores into a composite score.
[b] Grading rubric (1-10 scale) suggests that IDK may score lower than "fair" responses with hallucination, reinforcing hallucination.

shows that the expected score of offering an answer beats IDK (score 0) iff its confidence (i.e., probability of being correct) is $> t$.

Such penalties have been well-studied within hallucination research (Ji et al., 2023). However, we suggest two subtle variations which have statistical ramifications. First, we propose making the confidence threshold explicit in the instructions, whereas the prior work has largely omitted mentioning the confidence targets or penalties in the instructions. (A notable exception is the work of Wu et al. (2025) who introduce "risk-informing" prompts with explicit penalties.) The ideal penalty might reflect likely real-world harms, but that is impractical as it is specific to the problem, the target application, and the user group. Without transparent specification within the instructions, it would be difficult to achieve consensus among language-model creators on the correct thresholds. Similarly, students might bicker that grading is unfair given instructions that there is an unspecified penalty for errors. Instead, specifying confidence thresholds explicitly in each problem's instructions supports objective grading even if the specific thresholds chosen are somewhat arbitrary or even random. A single model may be best across all thresholds, if the threshold is explicit. However, if the threshold is not stated, then there is an inherent tradeoff, and no single model will be best in general (other than one that is always correct).

Second, we suggest incorporating confidence targets into existing mainstream evaluations, such as the popular SWE-bench (Jimenez et al., 2024) which involves binary grading of software patches, while the majority of prior work has introduced implicit error penalties in bespoke hallucination evaluations. Merely adding evaluations with implicit error penalties faces the aforementioned accuracy-error tradeoff. On the other hand, incorporating confidence targets into the established evaluations, already in use, reduces the penalty for appropriate expressions of uncertainty. It may thus amplify the effectiveness of hallucination-specific evaluations.

With explicit confidence targets, there is one behavior which is simultaneously optimal for all targets—outputting IDK among examples where its correctness probability is greater than the

target. Let us refer to this as *behavioral calibration*–rather than requiring the model to output a probabilistic confidence (Lin et al., 2022a), it must formulate the most useful response in which it is at least t confident. Behavioral calibration can be audited by comparing accuracy and error rates across thresholds, and circumvents the problem that there may be exponentially many ways to phrase correct responses (Farquhar et al., 2024). Existing models may or may not exhibit behavioral calibration, but it may prove useful as an objective evaluation.

5 Discussion and limitations

It is difficult for the field to agree upon how to define, evaluate and reduce hallucinations due to their multifaceted nature. A statistical framework must prioritize certain aspects and omit others, for simplicity. Several notes are in order about the extent and limitations of the framework used herein.

Plausibility and nonsense. A hallucination is a plausible falsehood, and by considering only plausible strings $\mathcal{X}$, our analysis ignores the possibility of generating nonsensical strings (which state-of-the-art language models rarely generate). However, the statement and proof of Theorem 1 hold with the modified definitions of nonsensical examples $\mathcal{N}$ with partition $\mathcal{X} = \mathcal{N} \cup \mathcal{E} \cup \mathcal{V}$, $\mathrm{err} := \hat{p}(\mathcal{N} \cup \mathcal{E})$, $D(\mathcal{N}) = 0$, and the assumption that $p(\mathcal{V}) = 1$.

Open-ended generations. For simplicity, the examples presented in this paper are oriented towards a single factual question. However, hallucinations often arise for open-ended prompts, such as "Write a biography about. . . ." This can be fit into our framework by defining a response containing one or more falsehoods to be an error. However, in such a case it would be natural to consider degrees of hallucination depending on how many errors there are.

Search (and reasoning) are not panaceas. A number of studies have shown how language models augmented with search or Retrieval-Augmented Generation (RAG) reduce hallucinations (Lewis et al., 2020; Shuster et al., 2021; Nakano et al., 2021; Zhang and Zhang, 2025). However, Observation 1 holds for arbitrary language models, including those with RAG. In particular, the binary grading system itself still rewards guessing whenever search fails to yield a confident answer. Moreover, search may not help with miscalculations such as in the letter-counting example, or other intrinsic hallucinations.

Latent context. Some errors cannot be judged by the prompt and response alone. For example, suppose a user asks a question about phones and the language model provides a response about cellphones, but the question was intended to be about land lines. Such ambiguities do not fit our error definition which does not depend on context external to the prompt and response. It would be interesting to extend the model to allow for "hidden context" that are not part of the prompt given to the language model, but which could be used for judging errors, relating to *aleatoric uncertainty*.

A false trichotomy. Our formalism does not distinguish between errors of different magnitudes or degrees of uncertainty. Clearly, the correct/incorrect/IDK categories are also incomplete. Although the statistical ideal might be to score each evaluation just as we would like to score the language model in the downstream application, explicit confidence targets offer a practical, objective modification to mainstream evaluations, and a false trichotomy may at least offer an IDK option unlike a false dichotomy.